

Financial Mathematics 32000

Lecture 8

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Numerical integration in Fourier space

Deriving a CF: Heston model

B-S implied vol in Heston model

Interview questions

Conclusions

(log) prices

payoff
probability density
Fourier transform

how strong

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be integrable, meaning $\int_{-\infty}^{\infty} |f(x)| dx < \infty$.

The *Fourier transform* of f is the function $\hat{f} : \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$\hat{f}(z) = \int_{-\infty}^{\infty} f(x) e^{izx} dx$$

frequency

 $i = \sqrt{-1}$

Theorem: If $\hat{f}$ is integrable then the *inversion* formula holds:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(z) e^{-izx} dz$$

(at every x where f is continuous)

Intuition: This represents $f(x)$ as a combination of functions e^{-izx}

having various “frequencies” z . The transform $\hat{f}(z)$ gives the

weighting of the e^{-izx} function. More precisely, ...

Intuition of inversion formula

skip

For arbitrary $L > 0$, on the interval $x \in [-\pi L, \pi L]$, the functions $e^{-ikx/L}$, for integer k , are orthonormal wrt the inner product

$$\langle a, b \rangle := \frac{1}{2\pi L} \int_{-\pi L}^{\pi L} a(x) \overline{b(x)} dx$$

General f may be represented on $[-\pi L, \pi L]$, using the “basis” functions $e^{-ikx/L}$. Specifically, the k th function has coefficient

$$\langle f, e^{-ikx/L} \rangle = \frac{1}{2\pi L} \int_{-\pi L}^{\pi L} f(x) e^{ikx/L} dx = \frac{1}{2\pi L} \widehat{f}_L(k/L).$$

where $f_L := f$ on $[-\pi L, \pi L]$, and 0 elsewhere. For $x \in [-\pi L, \pi L]$,

$$f(x) = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi L} \widehat{f}_L(k/L) e^{-ikx/L} \longrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(z) e^{-izx} dz$$

as $L \rightarrow \infty$.

Characteristic function

The characteristic function (CF) of any random variable X is the function $F_X : \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$F_X(z) := \mathbb{E}e^{izX}$$

If X has a density f , then $F_X(z) = \int_{-\infty}^{\infty} f(x)e^{izx}dx = \hat{f}(z)$, so the CF of X is the same thing as the Fourier transform of its density. We can also talk about the CF of a distribution, which means the same thing as the CF of a variable X with that distribution.

A characteristic function uniquely identifies a distribution.

Indeed, in the case that F_X is integrable, applying the inversion formula to F_X gives the density of X .

$$\mathbb{E} e^{i\tau(X+Y)} = \mathbb{E} e^{i\tau X} e^{i\tau Y}$$

Characteristic function

Just as a distribution can be specified by giving a CDF or a density, it can be specified by giving a CF.

Why may we like to work with CF instead of a density or a CDF?

- Sometimes the CF is much simpler than the density.
- The following facts help us prove limit theorems (such as CLT):
The CF of the *sum* of *independent* random variables is the *product* of the CFs. And the pointwise convergence of CFs to a continuous limit F is equivalent to the convergence of the corresponding distributions to a limiting distribution with CF F .
- Quantities of the form $\mathbb{E}g(X)$ can often be expressed easily in terms of the CF of X . This is useful for derivatives pricing.

Deriving a CF

Examples:

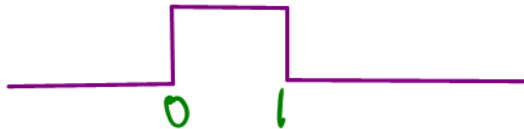
- ▶ Let X be Normal(a, b^2).

Then the CF of X is

$$F_X(z) = \mathbb{E}e^{izX} = e^{\mathbb{E}izX + (1/2)\text{Var}(izX)} = e^{iaz - b^2 z^2 / 2}$$

- ▶ Let U be Uniform[0, 1].

Then the CF of U is



$$F_U(z) = \mathbb{E}e^{izU} = \int_0^1 1 \times e^{izu} du = \left. \frac{e^{izu}}{iz} \right|_0^1 = \frac{e^{iz} - 1}{iz}$$

for $z \neq 0$. And $F_U(0) = 1$.

$\mathbb{E}e^{ix} = \mathbb{E}e^{i(-in)X} = F_X(-in)$

Using the CF to compute moments

The n th moment of X is the expectation $\mathbb{E}X^n$. To compute:

Take n derivatives of $F_X(z) = \mathbb{E}e^{izX}$ wrt z :

$$F_X^{(n)}(z) = \mathbb{E}((iX)^n e^{izX})$$

Evaluate at $z = 0$ to get $\mathbb{E}X^n = (-i)^n F_X^{(n)}(0)$.

► Example: If X is Normal(0, 1) then $F_X(z) = e^{-z^2/2}$ so

$$F_X'(z) = -ze^{-z^2/2} \Rightarrow \mathbb{E}X = (-i)(0) = 0$$

$$F_X''(z) = (-1 + z^2)e^{-z^2/2} \Rightarrow \mathbb{E}X^2 = (-i)^2(-1) = 1$$

To obtain moments of e^X : Let us extend the domain of F_X or $\hat{f}$ (which was $\mathbb{R}$) to a strip in $\mathbb{C}$ defined by $\{z : \mathbb{E}|e^{izX}| < \infty\}$.

Then $\mathbb{E}[(e^X)^n] = F_X(-in)$

Using the CF to compute the CDF

To obtain the CDF of X at any point of continuity k :

$$\mathbb{E}1_{X < k} = \mathbb{P}(X < k) = \frac{1}{2} - \frac{1}{\pi} \int_0^{\infty} \operatorname{Re} \left[\frac{F_X(z)}{iz} e^{-izk} \right] dz$$

Sanity check: the k -derivative of the RHS is

$$\begin{aligned} -\frac{1}{\pi} \int_0^{\infty} \operatorname{Re}[-F_X(z)e^{-izk}]dz &= \frac{1}{2\pi} \int_0^{\infty} F_X(z)e^{-izk} + F_X(-z)e^{izk} dz \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F_X(z)e^{-izk} dz = f(k), \end{aligned}$$

which recovers the density, as it should.

We have used the fact that

$$\overline{F_X(z)} = \overline{\mathbb{E}e^{izX}} = \mathbb{E}e^{-i\bar{z}X} = F_X(-\bar{z})$$

Using the CF to compute asset-or-nothing call price

Assume $\mathbb{E}e^X < \infty$. Define the measure $\mathbb{P}^*$ by the property that

$$\mathbb{E}^*Y = \mathbb{E}\left[\frac{e^X}{\mathbb{E}e^X}Y\right]$$

for all random variables Y such that the $\mathbb{E}$ exists. (If e^X is a time- T share price, then $\mathbb{P}^*$ is share measure.) The $\mathbb{P}^*$ -CF of X is

$$F^*(z) = \mathbb{E}^*e^{izX} = \mathbb{E}\left[\frac{e^X}{\mathbb{E}e^X}e^{izX}\right] = \frac{\mathbb{E}e^{(iz+1)X}}{\mathbb{E}e^X} = F_X(z-i)/F_X(-i).$$

Therefore, for $k \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}(e^X \mathbf{1}_{X>k}) &= \mathbb{E}e^X \mathbb{E}\left[\frac{e^X}{\mathbb{E}e^X} \mathbf{1}_{X>k}\right] = \mathbb{E}e^X \mathbb{P}^*(X > k) \\ &= F_X(-i) \left(\frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{F_X(z-i)/F_X(-i)}{iz} e^{-izk} \right] dz \right) \\ &= \frac{F_X(-i)}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{F_X(z-i)}{iz} e^{-izk} \right] dz.\end{aligned}$$

Using the CF to compute call prices

The time-0 price of a call on e^X struck at K is

$$e^{-rT} \left[\mathbb{E}(e^X \mathbf{1}_{e^X > K}) - K \mathbb{P}(e^X > K) \right]$$

Let $k := \log K$. The first term, by L8.18, equals

$$\mathbb{E}(e^X \mathbf{1}_{X > k}) = \frac{F_X(-i)}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{F_X(z-i)}{iz} e^{-izk} \right] dz$$

The second term, by L8.9, equals

$$K \mathbb{P}(X > k) = K \left(\frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{F_X(z)}{iz} e^{-izk} \right] dz \right)$$

Typical application: $X = \log S_T$.

This was the approach in Heston (1993).

Using the CF to compute $\mathbb{E}g(X)$ SKQ

Parseval/Plancherel Theorem: If f and g are integrable and square integrable ($\int_{-\infty}^{\infty} f^2 < \infty$ and $\int_{-\infty}^{\infty} g^2 < \infty$) then

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(z) \overline{\hat{g}(z)} dz = \left\langle \frac{\hat{f}}{\sqrt{2\pi}}, \frac{\hat{g}}{\sqrt{2\pi}} \right\rangle$$

Idea: The transformation $f \mapsto \hat{f}/\sqrt{2\pi}$ is *unitary*: it preserves inner products. Analogous to unitary mappings in $\mathbb{R}^n$: reflections/rotations that preserve lengths and angles (dot products) of vectors.

Application: Let $f :=$ density of $\log S_T$, and $g(\log S_T)$ be some payoff.

Example: for call option, $g(x) = (e^x - K)^+$. We want to compute

$$\mathbb{E}g(\log S_T) = \int_{-\infty}^{\infty} f(x) \overline{g(x)} dx = (\text{we hope}) \frac{1}{2\pi} \int \hat{f}(z) \overline{\hat{g}(z)} dz$$

But $\int (e^x - K)^+ dx = \infty$, so the call payoff g is not integrable.

Using the CF to compute $\mathbb{E}g(X)$

skip

Solution: let $f_\alpha(x) := e^{\alpha x} f(x)$ and $g_{-\alpha}(x) := e^{-\alpha x} g(x)$, where

$$1 < \alpha < \sup\{p : \mathbb{E}S_T^p < \infty\}.$$

Then $fg = f_\alpha g_{-\alpha}$ and

$$\hat{f}_\alpha(z) = \int_{-\infty}^{\infty} e^{\alpha x} f(x) e^{izx} dx = \int_{-\infty}^{\infty} f(x) e^{ix(z-\alpha i)} dx = \hat{f}(z - \alpha i)$$

and likewise $\hat{g}_{-\alpha}(z) = \hat{g}(z + \alpha i)$. Then

$$\begin{aligned} \mathbb{E}g(\log S_T) &= \int_{-\infty}^{\infty} f_\alpha(z) \overline{g_{-\alpha}(z)} dz = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(z - \alpha i) \overline{\hat{g}(z + \alpha i)} dz \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(z - \alpha i) \hat{g}(-z + \alpha i) dz = \frac{1}{2\pi} \int_{-\infty - \alpha i}^{\infty - \alpha i} \hat{f}(z) \hat{g}(-z) dz. \end{aligned}$$

(Note that $\hat{g}$ here is evaluated at points $u + \alpha i$ where $u \in \mathbb{R}$.)

Example: compute call price

skip

In the case that $g(x) := (e^x - K)^+$, the payoff transform is, at any point $z = u + \alpha i$ where $u \in \mathbb{R}$ and $\alpha > 1$,

$$\begin{aligned}\hat{g}(z) &= \int_{-\infty}^{\infty} (e^x - K)^+ e^{ixz} dx = \int_{\log K}^{\infty} (e^{x(iz+1)} - Ke^{ixz}) dx \\ &= \left(\frac{e^{(1+iz)x}}{1+iz} - K \frac{e^{izx}}{iz} \right) \Big|_{x=\log K}^{x=\infty} = K \frac{e^{iz \log K}}{iz} - \frac{e^{(1+iz) \log K}}{1+iz} = \frac{e^{(1+iz) \log K}}{z(i-z)}\end{aligned}$$

(the $x = \infty$ terms vanished because $1+iz$ and iz have real parts < 0).

$$\text{Hence } \mathbb{E}(e^X - K)^+ = \frac{1}{2\pi} \int_{-\infty - \alpha i}^{\infty - \alpha i} F_X(z) \frac{e^{(1-iz) \log K}}{-z(z+i)} dz.$$

Equivalently, via $z \mapsto z - i$, this is sometimes rewritten for $\alpha > 0$ as

$$\frac{1}{2\pi} \int_{-\infty - \alpha i}^{\infty - \alpha i} F_X(z - i) \frac{e^{-iz \log K}}{z(i-z)} dz$$

Computational benefits (vs. L8.11): One integral. Ability to choose α .

Numerical integration

SKX

Can rewrite this as an integral over positive real line. With $k = \log K$,

$$\begin{aligned}\mathbb{E}(e^X - K)^+ &= \frac{1}{\pi} \int_{0-\alpha i}^{\infty-\alpha i} \operatorname{Re} \left(F_X(z-i) \frac{e^{-izk}}{z(i-z)} \right) dz \\ &= \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left(F_X(x - (\alpha+1)i) \frac{e^{-i(x-\alpha i)k}}{(x-\alpha i)((\alpha+1)i-x)} \right) dx \\ &= \frac{1}{e^{\alpha k}} \int_0^\infty \operatorname{Re}(h(x)e^{-ixk}) dx\end{aligned}$$

where $h(x) = \frac{F_X(x-(\alpha+1)i)}{\pi(x-\alpha i)((\alpha+1)i-x)}$. Can evaluate integral using

- Fixed integration points, e.g. “midpoint” / “rectangular” rule

$$\Delta_x e^{-\alpha k} \sum_{n=1}^N \operatorname{Re}(h(x_n)e^{-ix_n k}) \text{ where } x_n = (n-1/2)\Delta_x$$

Can calculate this by simple summation, or by FFT.

- Adaptive integration algorithms, e.g. Python `scipy.integrate`

FFT

sk 70

FFT algorithm: Input $\mathbf{v} \in \mathbb{C}^N$. Output $\text{FFT}[\mathbf{v}] \in \mathbb{C}^N$ where

$$\text{FFT}[\mathbf{v}]_m = \sum_{n=1}^N e^{-i(2\pi/N)(n-1)(m-1)} \mathbf{v}_n, \quad m = 1, 2, \dots, N$$

Python `numpy.fft` computes FFT (with 0-based indexing). Can price calls at log-strikes $k_m = k_1 + (m-1)\Delta_k$ for $m = 1, \dots, N$, because

$$x_n k_m = (n-1)(m-1)\Delta_x \Delta_k + (n-1/2)k_1 \Delta_x + (m-1)\Delta_x \Delta_k / 2.$$

If $\Delta_x \Delta_k = 2\pi/N$ then running *one* FFT solves for *all* ($m = 1, \dots, N$) log-strikes k_m , by calculating the midpoint rule using FFT as

$$\Delta_x e^{-\alpha k_m} \text{Re} \sum_{n=1}^N h(x_n) e^{-i x_n k_m} = \Delta_x e^{-\alpha k_m} \text{Re}(e^{-i\pi(m-1)/N} \text{FFT}[\mathbf{v}]_m)$$

where $\mathbf{v}_n := h(x_n) e^{-i k_1 x_n}$.

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Steve Heston



The Heston (1993) stochastic volatility model

Let W^S and W^V be $\mathbb{P}$ -BM with correlation ρ . Let

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{V_t} S_t dW_t^S \\dV_t &= \kappa(\theta - V_t)dt + \eta\sqrt{V_t}dW_t^V\end{aligned}$$

Interpretation of the parameters $\kappa, \theta, \eta > 0$:

κ = rate of mean-reversion

θ = long-term mean

η = volatility of volatility

Then $X_t := \log S_t$ has dynamics

$$dX_t = \left(r - \frac{1}{2}V_t\right)dt + \sqrt{V_t}dW_t^S$$

The Heston CF

skip

We want to find the CF of X_T , in order to price options on S_T .

Let's find indeed, for all $t < T$, the time- t conditional CF

$$M_t := M(X_t, V_t, t; z) := \mathbb{E}_t e^{izX_T}$$

For each z , this M_t is a *martingale*, because if $s < t$ then

$$M_s = \mathbb{E}_s e^{izX_T} = \mathbb{E}_s [\mathbb{E}_t e^{izX_T}] = \mathbb{E}_s M_t.$$

So we *set its drift equal to zero*. By Itô $dM_t =$

$$\frac{\partial M}{\partial t} dt + \frac{\partial M}{\partial X} dX + \frac{\partial M}{\partial V} dV + \frac{1}{2} \frac{\partial^2 M}{\partial X^2} (dX)^2 + \frac{1}{2} \frac{\partial^2 M}{\partial V^2} (dV)^2 + \frac{\partial^2 M}{\partial V \partial X} (dV)(dX)$$

So we have a (backward Kolmogorov) PDE

$$\frac{\partial M}{\partial t} + \frac{\partial M}{\partial X} \left(r - \frac{1}{2}V\right) + \frac{\partial M}{\partial V} \kappa(\theta - V) + \frac{1}{2} \frac{\partial^2 M}{\partial X^2} V + \frac{1}{2} \frac{\partial^2 M}{\partial V^2} \eta^2 V + \frac{\partial^2 M}{\partial V \partial X} \rho \eta V = 0$$

Solve for A and B

skip

Let's guess that

$$M(X_t, V_t, t; z) = \mathbb{E}_t e^{izX_T} = e^{A(t; z) + izX_t + B(t; z)V_t}$$

for some A and B , which may depend on the model's parameters.

The guess is that the CF is exponential-*affine* wrt (X, V) . *Affine* means a constant plus a linear transformation. The PDE becomes

$$\frac{dA}{dt} + \frac{dB}{dt}V + iz(r - \frac{1}{2}V) + B\kappa(\theta - V) + \frac{1}{2}(iz)^2V + \frac{1}{2}\eta^2B^2V + izB\rho\eta V = 0$$

We want this to vanish for all V . So $A(T) = B(T) = 0$ and

$$\begin{aligned} \frac{dA}{dt} + irz + \kappa\theta B &= 0 \\ \frac{dB}{dt} - \frac{iz}{2} - \frac{z^2}{2} + (i\rho\eta z - \kappa)B + \frac{\eta^2}{2}B^2 &= 0 \end{aligned}$$

Solve for A and B

Skip

So

$$\frac{dB}{dt} = -\frac{\eta^2}{2}(B - c_1)(B - c_2)$$

where c_1, c_2 depend on ρ, η, κ, z . This is a *Riccati* ODE, with solution

$$B(t) = c_1 c_2 \frac{1 - e^{(T-t)(c_2 - c_1)\eta^2/2}}{c_1 - c_2 e^{(T-t)(c_2 - c_1)\eta^2/2}}.$$

The other ODE is $dA/dt = -irz - \kappa\theta B$ hence

$$A(t) = irz(T - t) + \kappa\theta \int_t^T B(u)du$$

which also has an explicit solution.

The Heston CF

Skip

Conclusion: The [time- t conditional] Heston CF is

$$F_X(z) = e^{A+izX_t+BV_t}.$$

$$A := irz(T-t) + \frac{\kappa\theta}{\eta^2} \left[(\kappa_* - \gamma)(T-t) - 2 \log \left(1 + \frac{\kappa_* - \gamma}{2\gamma} (1 - e^{-\gamma(T-t)}) \right) \right]$$

$$B := \frac{-(zi + z^2)(1 - e^{-\gamma(T-t)})}{2\gamma e^{-\gamma(T-t)} + (\gamma + \kappa_*)(1 - e^{-\gamma(T-t)})}$$

$$\kappa_* := \kappa - i\rho\eta z$$

$$\gamma := \sqrt{\kappa_*^2 + \eta^2(zi + z^2)}$$

Use the principal branch of the complex log with this formulation.

Plug $F_X(z)$ into L8.11 or L8.15 to get call prices.

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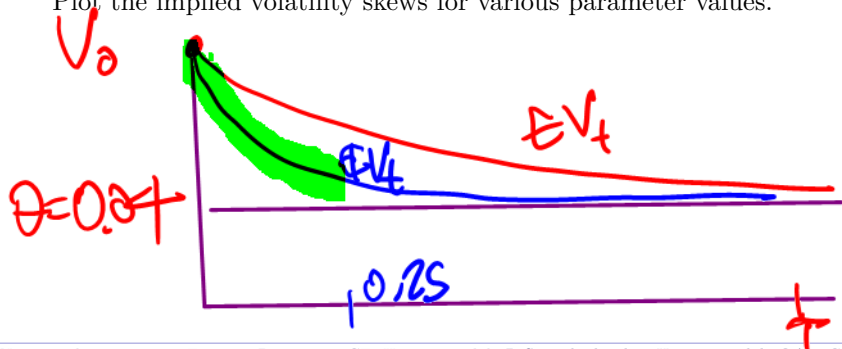
Heston model

Let $T = 0.25$.

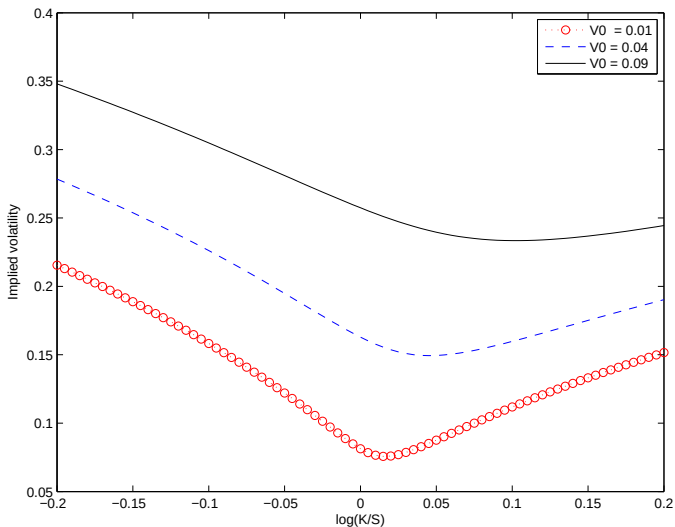
Let $r = 0$, $\kappa = 1$, $\theta = 0.04$, $\eta = 1.0$, $\rho = -0.5$, $V_0 = 0.04$.

Use Fourier method to compute Heston-model option prices, and express those prices as Black-Scholes implied volatilities.

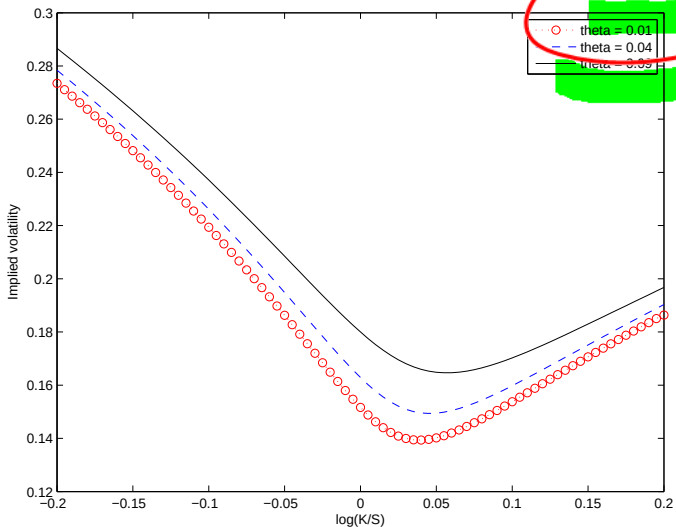
Plot the implied volatility skews for various parameter values.



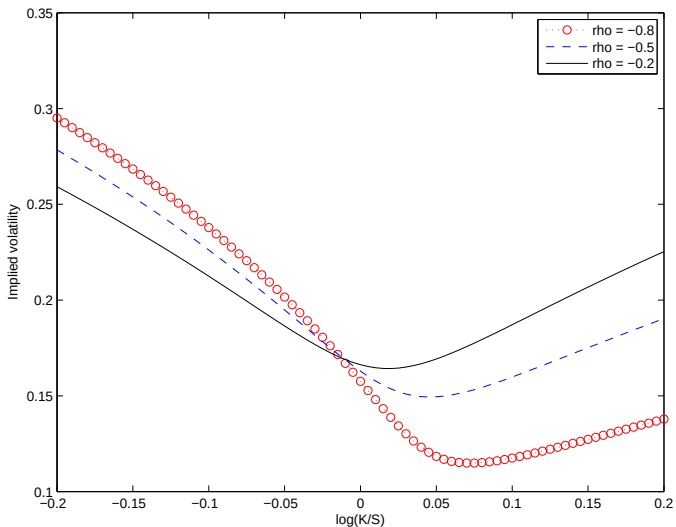
Effect of V_0



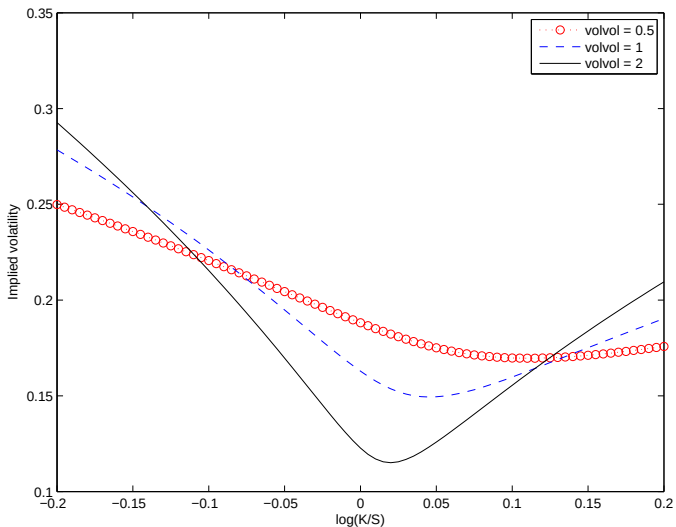
Effect of θ



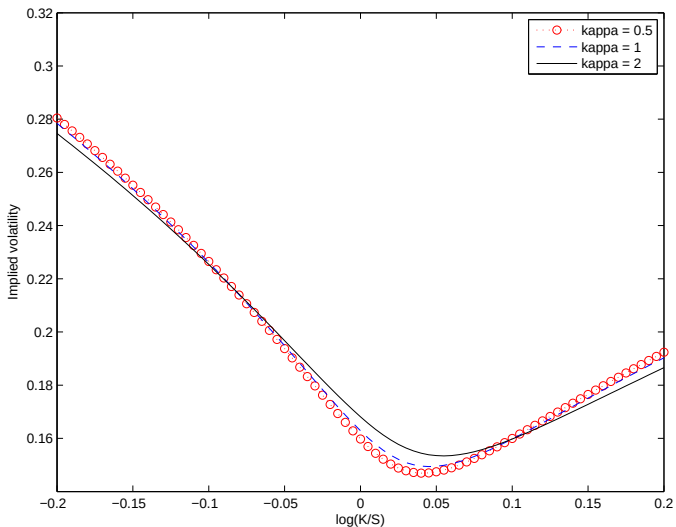
Effect of ρ



Effect of vol-of-vol η



Effect of κ



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Moment generating functions

- ▶ (Goldman) What is a moment generating function?

Answer: The moment generating function of X is

$$M_X(u) = \mathbb{E}e^{uX}$$

at all $u \in \mathbb{R}$ where this is finite. Similar to characteristic function, but CF always exists, while MGF may be infinite for all $u \neq 0$.

- ▶ (Goldman) Why is its first derivative the mean?

Answer: $M'_X(u) = \mathbb{E}Xe^{uX}$ so $M'_X(0) = \mathbb{E}X$.

- ▶ (FHLBC) How to calculate $\mathbb{E}X^2$ given the MGF of X ?

Answer: $M''_X(u) = \mathbb{E}X^2e^{uX}$ so $M''_X(0) = \mathbb{E}X^2$.

Heston

- ▶ (Murex) What is the Heston model?
- ▶ (Security Benefit) What are the parameters of the Heston model?
- ▶ (UBS) For Heston model, what will the volatility surface look like if the volatility of volatility increases?

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Numerical integration (quadrature)

- ▶ Given g the density of X , and payoff $f(X)$, numerical integration calculates the expectation $\int f(x)g(x)dx$.
- ▶ Midpoint quadrature (let $h = fg$, restrict to domain of length L , divided into N intervals of length $\Delta x = L/N$ and midpoints x_n . Numerical result: $\sum_n h(x_n)\Delta x$), is equivalent to doing first-order Taylor approximation of h at each midpoint.
Max error in approximating h on each interval is $O(\Delta x)^2$. Error in approximating $\int h$ on each interval is $O(\Delta x)^3$. Total error in approximating $\int h$ on N intervals is $N \times O(\Delta x)^3 = O(\Delta x)^2$.
- ▶ Thus one-dimensional numerical integration of payoff $\times$ density is a preferred approach, provided that the density is available.

Numerical integration in Fourier space

- ▶ Given the CF of X , another way to calculate expectations of payoffs $f(X)$ is by numerical integration in Fourier space.
Instead of integrating Payoff $\times$ Density, the Fourier transform approach integrates Payoff transform $\times$ Density transform (CF)
- ▶ One-dimensional numerical integration of payoff transform $\times$ CF is also a preferred approach – provided that the CF is available.
- ▶ Often useful for pricing vanilla contracts quickly – for example in *calibration* applications.

Trees and FD

Trees

- ▶ Trees can be regarded as *explicit* finite difference methods.
But FD have greater flexibility, because FD can also be done by implicit, C-N, etc.

Finite Differences

- ▶ Explicit: Simple. Equivalent to trinomial tree. Easier to parallelize. But only **first**-order accurate in Δt , and have stability restrictions.
- ▶ Implicit and C-N: Unconditionally stable and (in C-N case) **second**-order accurate in Δt . But requires solution of linear system at each time step.

Monte Carlo

- ▶ Easy to code (and *parallelize*) even for complex contract/dynamic
- ▶ Estimates have random noise, which goes to zero as $O(1/\sqrt{M})$.
- ▶ Advantages on multidimensional problems (Multi-asset contracts. Or multi-factor dynamics. Or some path-dependent contracts.)

For a 1-dimensional problem, FD methods typically more efficient than MC. But FD computational burden grows exponentially as the number of dimensions grows. (“Curse of Dimensionality”).

- ▶ FD / quadrature: If error is $\text{constant}/N^2$ and work is $\text{const} \times N^D$ then work to achieve ε error is $\text{const} \times (\text{constant}/\sqrt{\varepsilon})^D$.

MC: standard error $\sqrt{\text{Var } Y}/\sqrt{M}$. If work is $\text{constant} \times DM$

then work to achieve ε standard error is $\text{constant} \times D \text{Var}(Y)/\varepsilon^2$,

which is “worse” than FD if $D < 3$.

Conclusions

Fastest to slowest execution:

- ▶ Explicit formula, such as Black-Scholes
- ▶ 1-dimensional numerical integration
- ▶ Low-dimensional PDE solution, or numerical integration, or trees
- ▶ Monte Carlo
- ▶ High-dimensional PDE solution, or numerical integration, or trees

But remember:

- ▶ Ease of coding may be relevant. Parallelizability may be relevant.
- ▶ Finance rewards those who see relevant relationships/similarities between A and B (which may denote assets/risks/situations). For pricing purposes, this $\rightarrow$ simplifications/approximations/bounds.